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Jean-Marc Jézéquel & Benoit Combemale (IRISA/DiverSE)
Antoine Beugnard (IMT-Atlantique)

Subject: Application for PhD Position – Composing DTs with Variability, Fidelity and Uncertainty

Dear Jean-Marc Jézéquel, Benoit Combemale, and Antoine Beugnard,

I am applying for the PhD position on "Composing DTs with Variability, Fidelity and Uncertainty" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

In my work on the Industrial Gas Turbine Digital Twin, I faced the precise challenge of managing fidelity and uncertainty. Rather than relying on exhaustive telemetry, I orchestrated an Edge-TCN pipeline that intelligently sampled data, balancing computational overhead with physical fidelity. This hands-on experience managing architectural variability and quantifying uncertainty in dynamic cyber-physical systems positions me perfectly to formalize these concepts across the DiverSE team's platforms.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui